

Class Test 5 (28 September 2026)

- **Definitions:** Let x_0 be a point in the domain of $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with $f(x_0) = b_0$.
 - f is **locally onto** at x_0 if there is an open ball $B_r(x_0)$ about x_0 in $\mathbb{R}^n$ and an open ball $B_s(b_0)$ about b_0 in $\mathbb{R}^m$ such that for *every* b in $B_s(b_0)$ there is *at least one* x in $B_r(x_0)$ such that $f(x) = b$.
 - f is **locally one-to-one** at x_0 if there is an open ball $B_r(x_0)$ about x_0 in $\mathbb{R}^n$ and an open ball $B_s(b_0)$ about b_0 in $\mathbb{R}^m$ such that for *every* b in $B_s(b_0)$ there is *at most one* x in $B_r(x_0)$ such that $f(x) = b$.
- (a) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a C^1 function with $f(x^*) = b^*$. Let $Df(x^*)$ denote the $m \times n$ Jacobian matrix of f at x^* .
 - (i) Prove that if $\text{rank}(Df(x^*)) = m \leq n$, then f is locally onto at x^* .
 - (ii) Prove that if $\text{rank}(Df(x^*)) = n \leq m$, then f is locally one-to-one at x^* .
- (b) Prove the following theorem.

(Inverse Function Theorem) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a C^1 function with $f(x^*) = y^*$. If $Df(x^*)$ is nonsingular then there exists an open ball $B_r(x^*)$ about x^* and an open set V about y^* such that f is a one-to-one and onto map from $B_r(x^*)$ to V . The inverse map $f^{-1} : V \rightarrow B_r(x^*)$ is also C^1 and $Df^{-1}(f(x^*)) = (Df(x^*))^{-1}$.